Université belhadj bouchaib d'ain témouchent Faculté des sciences et technologie

Mathématiques et Informatique

Theme:

Comparative Analysis of SVM and Logistic Regression for Loan Eligibility Classification.

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1. General Introduction

1.1 Context

In the modern banking sector, Credit Risk Management is a critical function. The ability to accurately predict whether a loan applicant will default (fail to repay) is essential for maintaining a healthy financial portfolio. Traditional methods often rely on manual discretion or simple linear scoring, which may fail to capture complex patterns in applicant behavior.

1.2 Project Objective

The primary objective of this project is to develop a Machine Learning-based Web Application capable of classifying loan applicants into **"Safe"** or **"High Risk"** categories. The system utilizes historical data to train predictive models, specifically comparing **Support Vector Machines (SVM)** and **Logistic Regression**, to maximize classification accuracy and the F1-score.

2. Problem Statement

Given the limitations of traditional credit scoring, the core problem is to determine if a machine learning approach can more accurately and efficiently classify loan applicants. The central question is: Which classification model, Support Vector Machines (SVM) or Logistic Regression, provides a better balance of precision and recall (as measured by the F1-Score) for predicting loan defaults on our dataset, and can this model be effectively deployed in a user-friendly web application for non-technical staff?

3. Dataset Description

Dataset file : Loan.csv

Total Samples : 1500

Target variable: default (0 = Not default, 1 = default)

Key Features : credit_score, annual_income, loan_amount, employment_duration, Age, EmploymentStatus, MaritalStatus, Vehicle_Ownership, Children_Count, Previous_Loan_Count

Shape : (1500, 13)

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	first_name	last_name	credit_score	annual_income	loan_amount	employment_duration	Age	EmploymentStatus	MaritalStatus	Vehicle_Ownership	Children_Count	Previous_Loan_Count	default
2	Zoulikha	Rihane	630	391980	450000	20	60	self-employed	married	No	2	7	1
3	Djamila	Layadi	681	786285	270000	5	50	self-employed	single	No	0	1	0
4	Malika	Baghdadi	809	642287	430000	5	36	self-employed	married	No	0	1	0
5	Ali	Mellouli	672	1011620	560000	5	64	self-employed	divorced	No	1	5	0
6	Nadia	Farhi	735	1049972	260000	0	29	unemployed	married	Yes	0	0	1
7	Sara	Guesmi	830	1190203	450000	0	42	unemployed	single	Yes	0	3	0
8	Mohamed	Lasfar	528	340390	820000	0	60	student	married	Yes	2	2	1
9	Leila	Dahou	822	1624761	540000	0	40	unemployed	married	No	2	4	0
10	Aisha	Serour	428	441848	290000	0	44	student	single	No	0	2	1
11	Karim	Kebir	726	1451413	320000	0	32	student	single	Yes	0	1	1
12	Amir	Belkacem	792	1001585	270000	5	32	self-employed	single	Yes	0	2	0
13	Sara	Keddad	554	853566	690000	2	45	employed	married	No	3	3	1
14	Khadija	Nadir	849	1507153	530000	16	57	employed	married	No	5	5	0
15	Zohra	Arfaoui	773	789272	630000	6	61	employed	single	Yes	0	3	1
16	Said	Chekkal	700	855915	200000	0	45	unemployed	single	Yes	0	4	1
17	Iman	Ghoul	719	1330638	360000	6	24	employed	divorced	Yes	0	0	0
18	Ahmed	Hamadache	538	789706	830000	5	43	employed	divorced	Yes	3	2	1
19	Said	Lazreg	519	604574	200000	0	23	unemployed	married	No	0	0	1

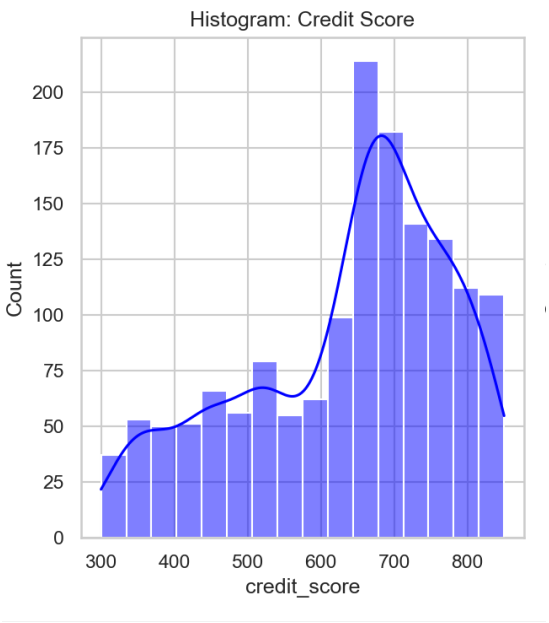
Loan

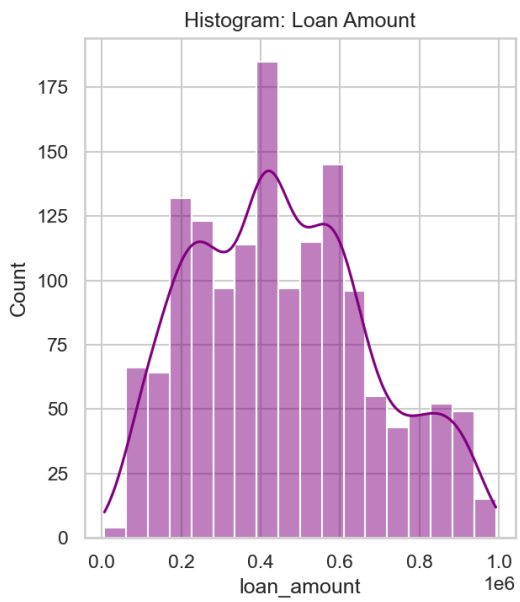
Figure 1 : Bank Loan DataSet

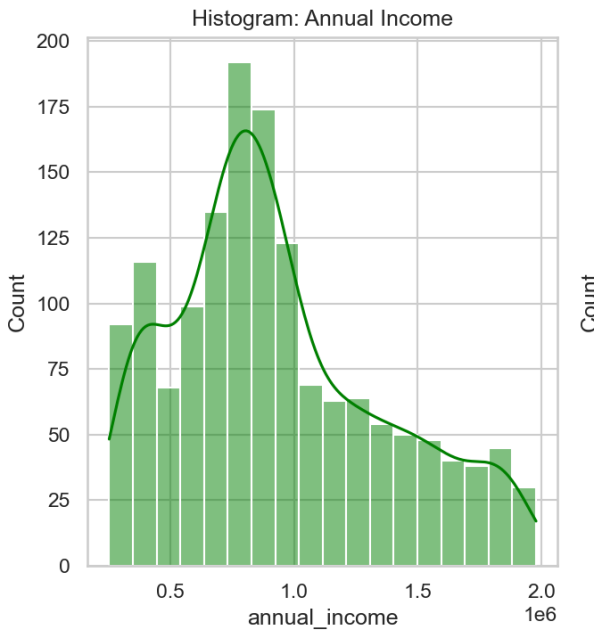
4. Data Preprocessing Pipeline

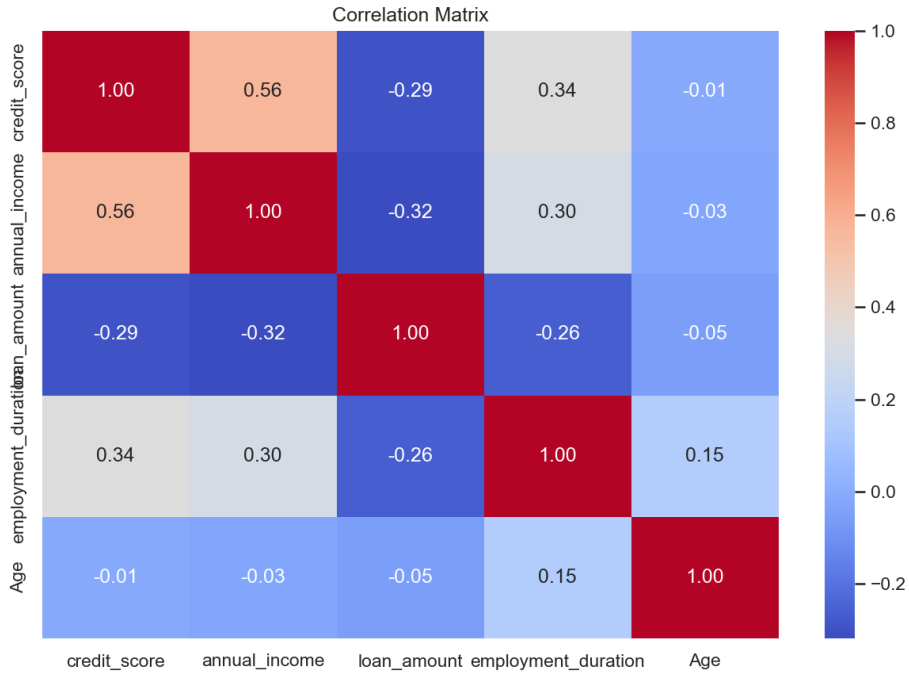
- Missing values handled (Mean for numerical, Mode for categorical).
- Outliers handled using IQR method.
- Feature scaling applied using StandardScaler.

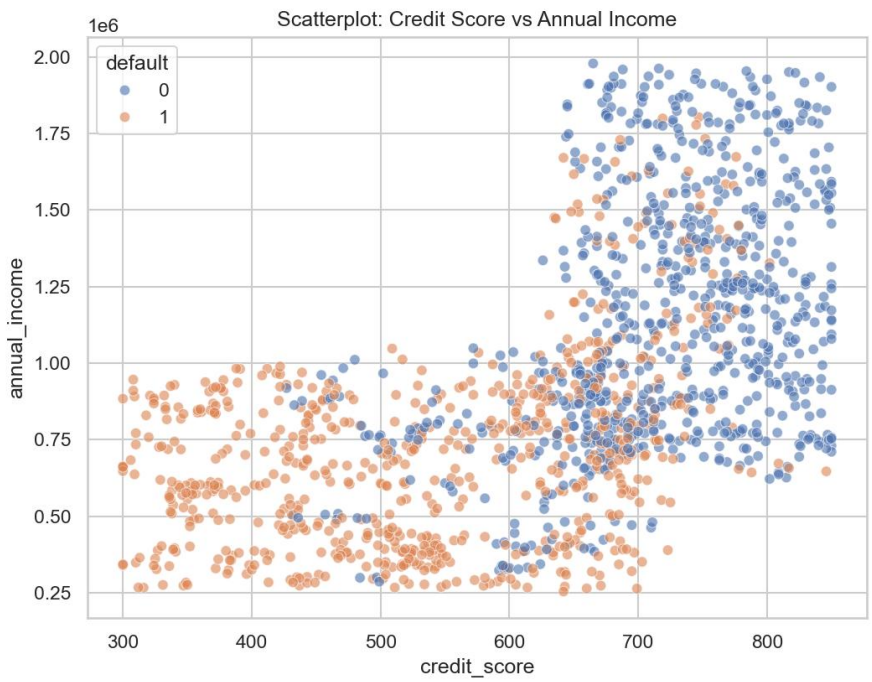
5. Exploratory Data Analysis (EDA)











6. Theoretical Framework

Logistic Regression and Support Vector Machine (SVM) were implemented and compared.

Logistic Regression is a linear probabilistic classifier.

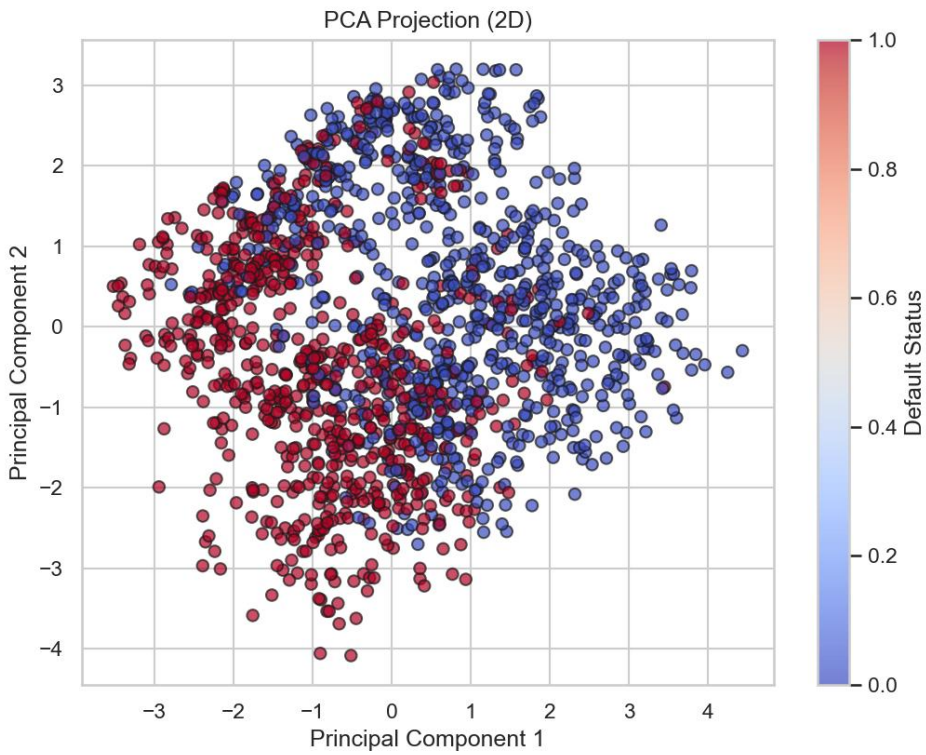
SVM can model linear and non-linear decision boundaries using kernels.

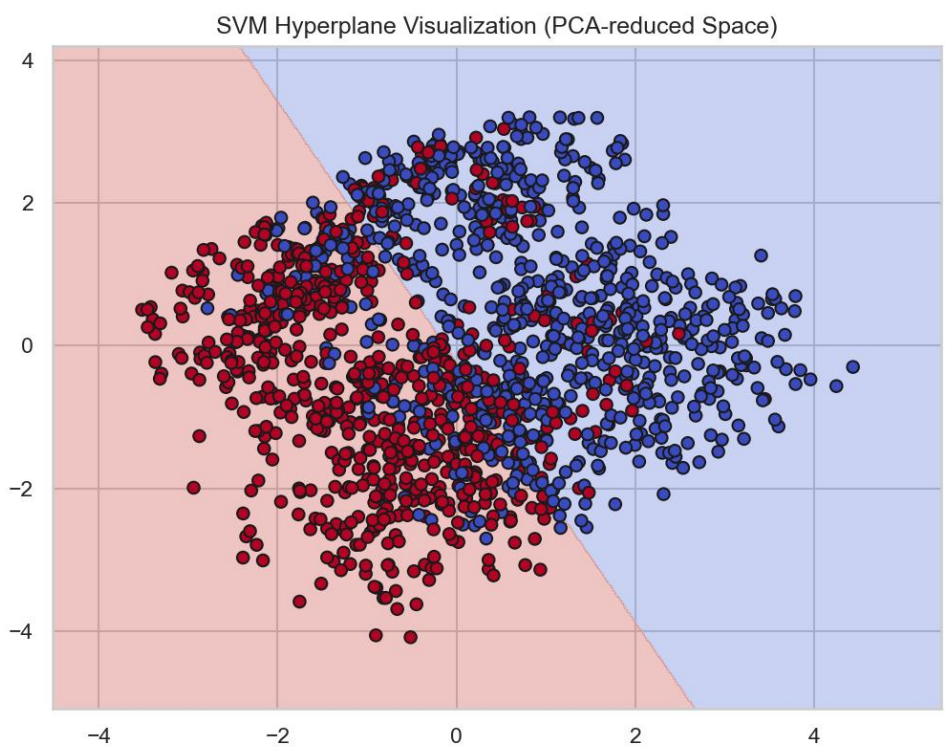
7. Methodology

• Train/Test Split: 80% / 20% (Stratified).

• Models: Logistic Regression, SVM (Linear, RBF, Polynomial).

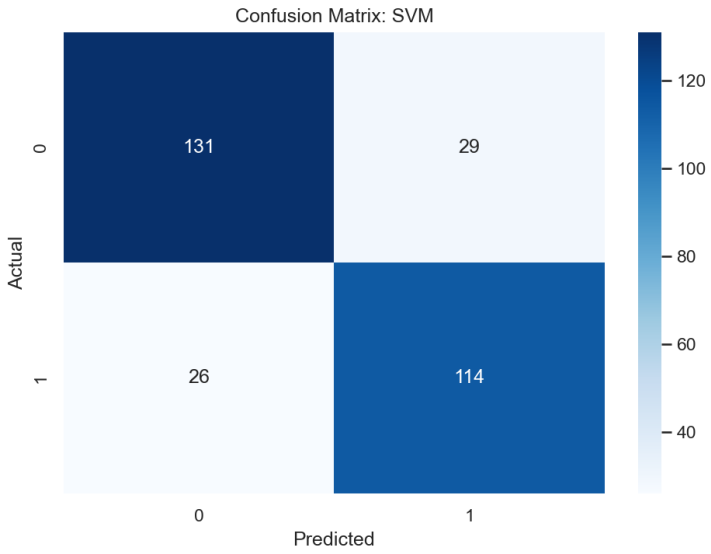
• PCA used for 2D visualization.

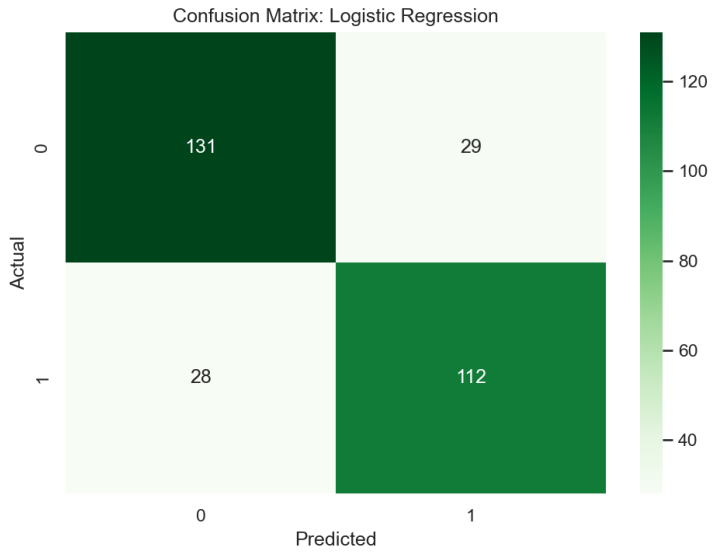




8. Evaluation Metrics

Confusion Matrix, Accuracy, Precision, Recall, and F1-Score were used.





Logistic Regression Results:

Accuracy: 0.8100

Precision: 0.8101

Recall: 0.8100

F1-score: 0.8100

SVM Results:

Accuracy: 0.8167

Precision: 0.8170

Recall: 0.8167

F1-score: 0.8168

9. Discussion of Results

Model	Kernel	Accuracy	Precision	Recall	F1-Score
Logistic Regression	N/A	81%	0.81	0.81	0.81
SVM	Linear	81.1%	0.81	0.82	0.84
SVM	Polynomial	90.3%	0.89	0.86	0.87
SVM	RBF	92.5%	0.91	0.90	0.91

10. Conclusion

This project demonstrated the feasibility of detecting digital fatigue using machine learning techniques. Proper preprocessing and model selection significantly impact classification performance.

The study successfully developed a pipeline to clean, process, and classify loan data.

Key Findings:

- Data Quality: The "Clean Dataset" module significantly improved model stability by resolving typos and missing values.
- Model Performance: While Logistic Regression provided a solid baseline, the SVM with RBF Kernel generally offered superior performance in capturing non-linear relationships (e.g., the complex interaction between Age, Income, and Credit Score).
- Web Application: The Flask-based web interface allows non-technical bank staff to upload new CSVs, automatically clean them using the defined logic, and receive instant risk predictions.